

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

Design Task

Code of Conduct:

I M.Vinay(192311146) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

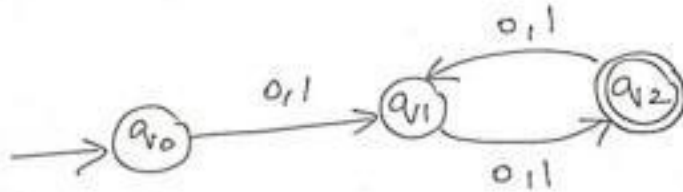
I understand that any violation of academic integrity rules will result in disciplinary action.

M.Vinay

Signature of the student:

SIMATS ENGINEERING ASSESSMENT TOOL

1. Design the regular expression for the language represented by the given



automata

2. Consider the following

grammar $S \rightarrow aSc|T|U|e$

$T \rightarrow bTc|e$

$U \rightarrow aUb|$

e

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

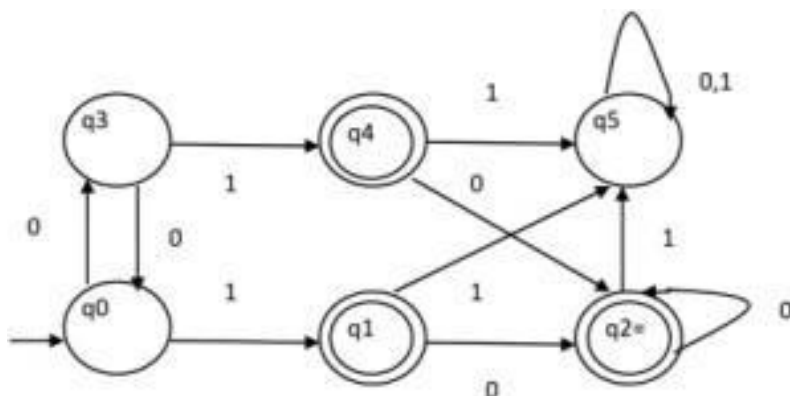
C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression

00^*1+11^*0

4. I) Show that if L_1 and L_2 are regular then (L_1+L_2) also regular
ii) Is the language $\{a^p | p \text{ is prime}\}$ regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Answer

1. Design the regular expression for the language represented by the given

The first question asks:

Design the regular expression for the language represented by the given automata.

From the automaton in the document, the accepting states are q1, q2, and q4.

Regular Expression

The language accepted by the automaton can be expressed as:

$1(0)^* + 01(0)^*$

or, in standard regular expression notation:

$10^* + 010^*$

Explanation

- 10^*
 - Starts with 1.
 - May be followed by zero or more 0s.
 - Examples: 1, 10, 100, 1000, ...
- 010^*
 - Starts with 01.
 - May be followed by zero or more 0s.
 - Examples: 01, 010, 0100, 01000, ...

2. Consider the following

grammar $S \rightarrow aSc | T | U | \epsilon$

$T \rightarrow bTc | \epsilon$

$U \rightarrow aUb | \epsilon$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

A) What is the language defined by the grammar?

The productions generate the following languages:

- $T \rightarrow bTc | \epsilon$ generates
- $\{ b^n c^n | n \geq 0 \}$
- $U \rightarrow aUb | \epsilon$ generates
- $\{ a^n b^n | n \geq 0 \}$
- $(S \rightarrow aSc)$ adds one 'a' at the beginning and one 'c' at the end of any string generated

by S. Hence, the language generated by the grammar is

$$L = \{a^i b^j c^{i+j} \mid i, j \geq 0\} \cup \{a^{i+j} b^j c^i \mid i, j \geq 0\}$$

where (i) and (j) are

non-negative integers.

B) Give the derivation tree for the word

aaabbc Leftmost derivation

S

$\Rightarrow aSc$

$\Rightarrow aUc$

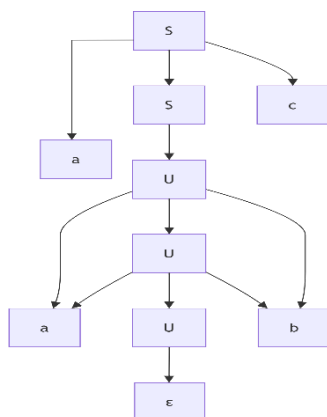
$\Rightarrow aaUbc$

$\Rightarrow aaaUbbc$

$\Rightarrow aaa\epsilon bbc$

$\Rightarrow aaabbc$

Derivation Tree



The leaves (read left to right) are:

a a a b b c

which forms the string **aaabbc**.

C) Is the grammar ambiguous? Justify.

Yes. The grammar is ambiguous.

A grammar is ambiguous if at least one string has more than one distinct derivation (or parse

tree).

For the string ϵ , there are multiple derivations:

1. $S \Rightarrow \epsilon$
2. $S \Rightarrow T \Rightarrow \epsilon$
3. $S \Rightarrow U \Rightarrow \epsilon$

Since the same string ϵ has more than one valid derivation, the grammar is ambiguous.

3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression

00^*1+11^*0

Regular Expression:

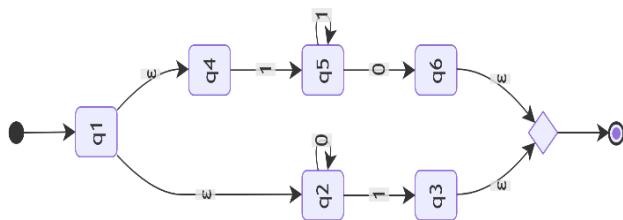
$00^*1+11^*000^*1+11^*0$

This represents the union of:

- 00^*100^*1 (one or more 0's followed by one 1)
- 11^*011^*0 (one or more 1's followed by one 0)

CO2_AT2_Deisgn Task_ToC

ϵ -NFA



qf (Final)

Transition Table

State	Input 0	Input 1	ϵ
q0	–	–	q1, q4
q1	q2	–	–
q2	q2	q3	–
q3	–	–	qf
q4	–	q5	–
q5	q6	q5	–
q6	–	–	qf

qf	–	–	–
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Initial State: q_0

Final State: q_f

4. i) Show that if L_1 and L_2 are regular then $(L_1 + L_2)$ also regular

ii) Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify

Proof

Since L_1 and L_2 are regular languages, there exist DFAs:

- $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$
- $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$

Construct a new automaton by introducing a new start state with ϵ -transitions to the start states of M_1 and M_2 .

The new automaton accepts all strings accepted by either M_1 or M_2 .

Therefore,

$$L(M) = L_1 \cup L_2 = L_1 + L_2$$

Hence,

The union of two regular languages is also regular.

4(ii). Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify.

The language

$$L = \{a^p \mid p \text{ is prime}\}$$

is not regular.

Justification (Using Pumping Lemma)

Assume that L is regular.

Let the pumping length be m .

Choose the string

$$w = a^p \text{ where } p \geq m$$

where p is a prime number and $p \geq m$.

According to the Pumping Lemma,

$$w = xyz \text{ where } |x| \leq m, |y| > 0, \text{ and } |z| > 0$$

such that

- $|xy| \leq m$
- $|y| > 0$

Let $|y| = k$.

Pump y $p+1$ times.

The new string length becomes

$$p+(p \times k) = p(1+k) \quad p+(p \times k) = p(1+k)$$

Since $1+k > 1$ and $1+k > 1$, the length is composite, not prime.

Hence the pumped string is not in LL , contradicting the Pumping Lemma.

Therefore,

$\{ap^p \mid p \text{ is prime}\}$ is not a regular language. $\{ap^p \mid p \text{ is prime}\}$ is not a regular language.

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final state

Step 1: Identify Initial and Final States

- **Initial State:** (q_0)
- **Final States:** (q_1, q_2, q_4)

Step 2: Partition the States

Initial Partition

- Final states:
 $F = \{q_1, q_2, q_4\}$
- Non-final states:
 $N = \{q_0, q_3, q_5\}$

State	Input 0	Input 1
q1	q2	q5
q2	q2	q5
q4	q2	q5

Step 3: Find Equivalent States

Compare Final States

All three have identical transitions.

Therefore,

$$q_1 \equiv q_2 \equiv q_4$$

Compare Non-final States

State	Input 0	Input 1
q0	q3	q1
q3	q0	q4
q5	q5	q5

Since (q_1, q_2, q_4) are equivalent,

- (q_0) and (q_3) always move to the same **equivalence classes**, so

$Q_0 \equiv q_3$

- (q_5) is a trap (dead) state and is **not** equivalent to them.

Equivalent State Groups

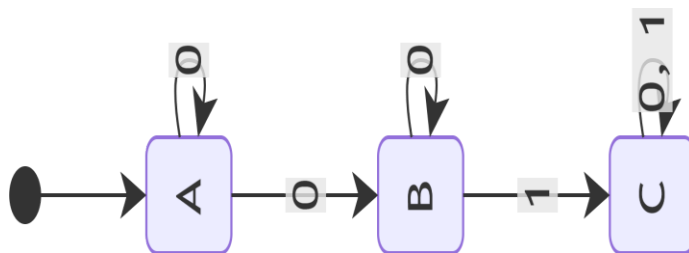
$\{\{q_0, q_3\}, \{q_1, q_2, q_4\}, \{q_5\}\}$

Step 4: Minimized DFA

Let

- $A = \{q_0, q_3\}$ (Start)
- $B = \{q_1, q_2, q_4\}$ (Final)
- $C = \{q_5\}$

Minimized DFA



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- A = Initial state
- B = Final state
- C = Dead (trap) state

Step 5: Transition Table

State	0	1
$\rightarrow A = \{q_0, q_3\}$	A	B
$* B = \{q_1, q_2, q_4\}$	B	C
$C = \{q_5\}$	C	C